

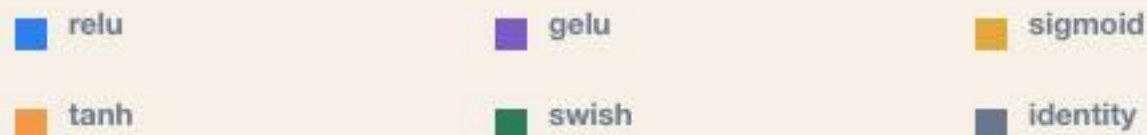
Activation Function Layouts

Simulated Annealing searches the activation assignments; gradient descent trains the weights.

What is optimized?

Weights: continuous parameters learned by gradient descent.

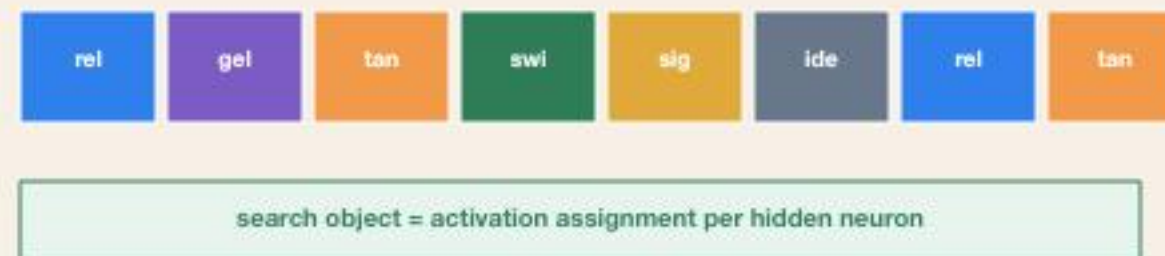
Layout: discrete activation choices searched by SA.



Fixed topology



Layout vector for one hidden layer



Short-Retrain Scores a Candidate After a Small Training Run.

This path is intuitive: generate a layout, train it briefly, then score it on validation loss.



Short Training Mixes Layout Quality With Training Noise.

A candidate can look good or bad for reasons that are not caused by the activation layout itself.

$$\text{candidate score} = \text{layout effect} + \text{training noise}$$

Random weights

A fresh initialization can dominate the score when the training budget is short.

Mini-batch sample

A small batch may favour or punish a candidate by chance.

Training duration

Too few updates may not integrate useful activation changes.

Optimizer noise

Measured improvement can reflect a lucky gradient step.

The next method keeps the comparison controlled: same weights, same mini-batch, only the layout changes.

Online-Delta Changes Only the Layout During Candidate Evaluation.

Weights and mini-batch stay fixed, so the loss difference isolates the immediate effect of the AF change.



Current layout



Candidate layout



only one AF assignment changed; weights are not reinitialized

The Candidate Is Judged by a Controlled Loss Difference.

The same batch and same weights make the before/after comparison meaningful.



$$\Delta = \text{loss}(\text{candidate}) - \text{loss}(\text{current})$$

$\Delta \leq 0$ means the candidate is better or equal on this mini-batch. $\Delta > 0$ means it is worse, but SA may still accept it.

Simulated Annealing Converts Loss Delta Into Exploration.

The temperature decides how willing the search is to keep worse candidates.

$$p(\text{accept worse}) = \exp(-\Delta / T)$$



High temperature

More exploration. Worse layouts can survive, so the search can escape local minima.

Low temperature

More conservative. The search increasingly prefers immediate loss improvements.

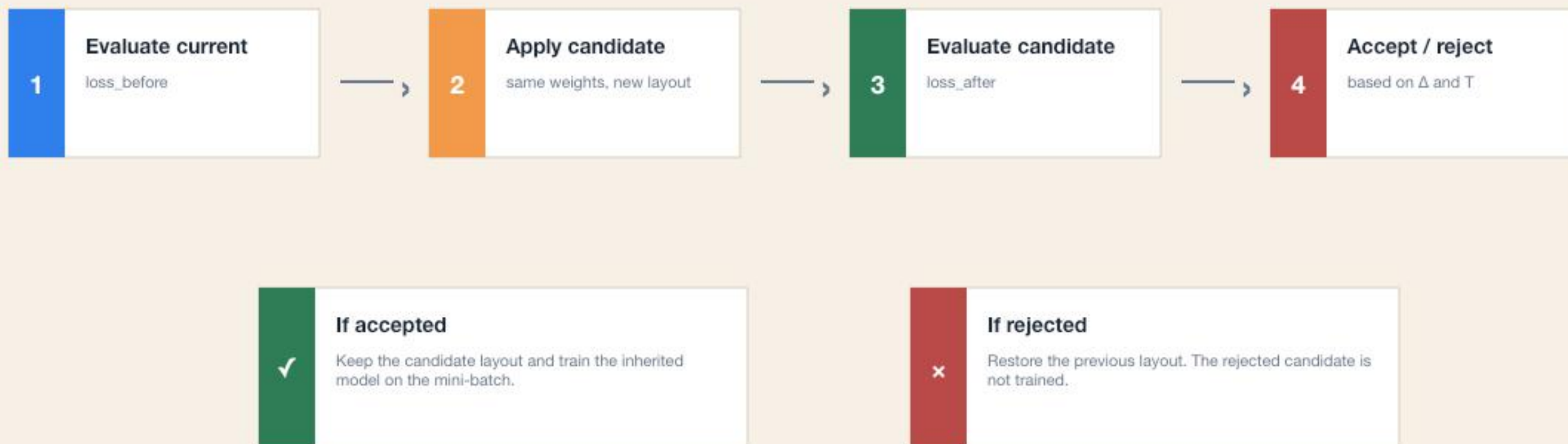
Cooling schedule

Temperature decreases over steps, shifting from exploration toward exploitation.



Training Happens After the Acceptance Decision.

This is the practical difference: the candidate is not trained first to make it look better.

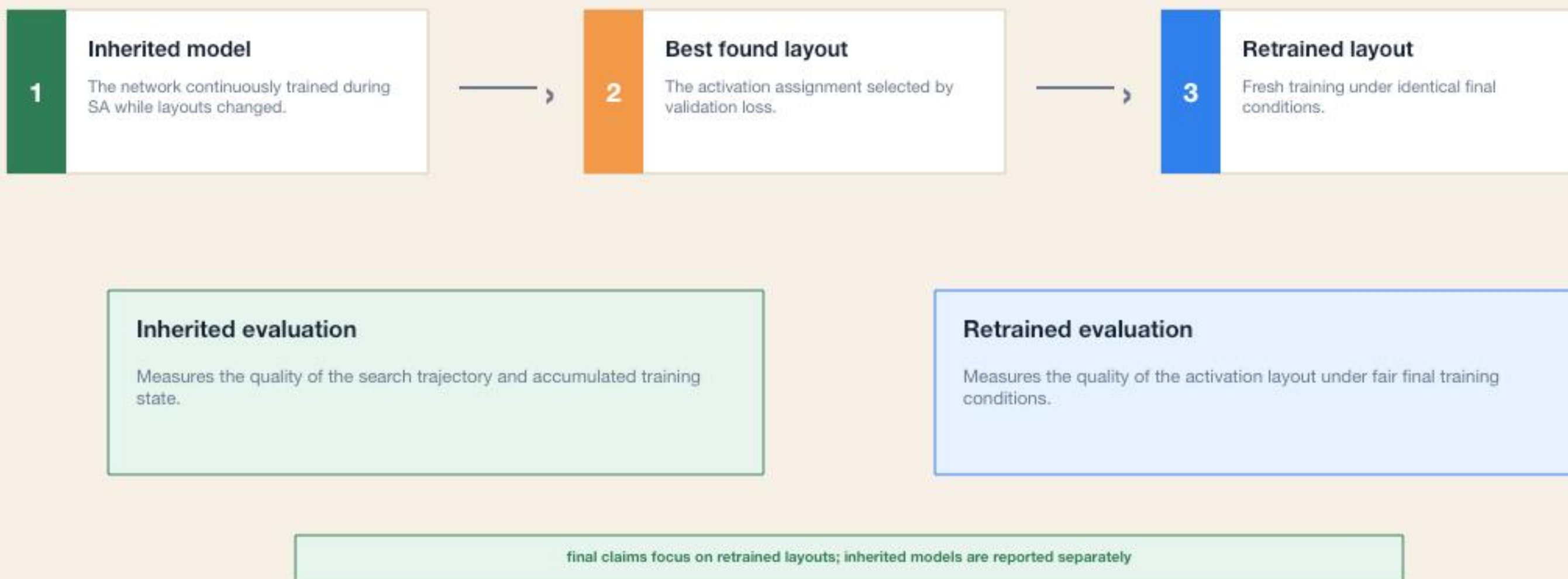


evaluate → accept/reject → train only if accepted

This order preserves the candidate evaluation as a layout test, not a short-training contest.

Inherited Networks and Retrained Layouts Answer Different Questions.

Online-Delta produces a trained search state, but fair layout claims require retraining selected layouts.



Neighborhood Operations Define the Local Layout Moves.

SA does not choose arbitrary networks; it proposes small, controlled changes around the current layout.



operation choice controls how locally or aggressively the search explores

The Evaluation Frame Is Fixed Before Comparing Layouts.

Search uses loss; test data is reserved for final reporting.

Benchmark	Topology	Baseline training	Tuning range	Role
concentric_circles	2-8-1	100 epochs	fixed first	easy sanity check
iris	4-8-3	150 epochs	up to 500	medium multi-class benchmark
crossing_spirals	6-16-16-1	250 epochs	up to 500	hard stress case

activation set: relu | gelu | sigmoid | tanh | swish | identity

Common reporting metrics

mean validation loss • std validation loss • mean test loss • mean test accuracy

E1-E7 Turn the Method Into a Fair Comparison.

The baselines answer whether SA adds value beyond obvious or random layouts.

ID	Experiment	Layouts / starts	Seeds	Purpose
E1	single-AF baselines	all_relu, all_gelu, all_sigmoid, all_tanh, all_swish, all_identity	10	strong non-SA reference
E2	random mixed baseline	random activation layouts	10	checks whether random mixing is already competitive
E3	online-delta SA	start from all_relu	10	can SA improve a common baseline?
E4	online-delta SA	start from all_tanh	10	can SA improve another smooth baseline?
E5	online-delta SA	start from all_sigmoid	10	can SA escape a weak saturating baseline?
E6	online-delta SA	start from all_identity	10	does SA damage or improve a strong simple layout?
E7	online-delta SA	start from random layout	10	does SA refine an already mixed layout?

final comparison: start layout vs best SA layout vs end SA layout vs E1/E2 baselines

A Useful SA Method Must Improve Layouts Under Fair Final Training.

The claim is not based on one search trajectory, but on retrained multi-seed comparisons.

1

Primary ranking

mean validation loss across seeds

2

Stability

standard deviation across seeds

3

Final reporting

test loss and test accuracy

Success criterion

Online-Delta is useful if it improves weak start layouts and remains competitive with simple baselines.

Open ablations

Compare short-retrain vs online-delta and test proposal-operation probabilities after E1-E7 are stable.

interpretation boundary: no superiority claim without mean/std over the full experiment matrix